Applied Algebra

Spring 2026

Instructor: Tuan Tran

Exercise Sheet 2

Deadline: by the end of next Wednesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week. Each week one person should write the solutions, and each student should do so for at least five weeks during the semester.

Please submit written solutions to **TWO** of the following problems.

Problem 1.

Find the following inverses, if they exist:

(i) the inverse of 7 modulo 11;

(ii) the inverse of 10 modulo 26;

(iii) the inverse of 11 modulo 31;

(iv) the inverse of 23 modulo 31;

(v) the inverse of 91 modulo 237.

Problem 2. Two players alternately add 1, 2, or 3 to a running total, starting from 0. The player who first reaches the target n wins.

Determine the winner and a winning strategy for:

(i) $n = 100$;

(ii) $n = 2026$.

Problem 3.

(i) Let a, b, n be integers with $n > 1$. Show that if $\gcd(a, n) = \gcd(b, n) = 1$, then $\gcd(ab, n) = 1$.

(ii) Let $n_1, \ldots, n_k$ be pairwise coprime positive integers, and let $a_1, \ldots, a_k$ be arbitrary integers. Show that the system of congruences

$$x \equiv a_1 \pmod{n_1}, \quad x \equiv a_2 \pmod{n_2}, \quad \ldots, \quad x \equiv a_k \pmod{n_k}$$

has a unique solution modulo $n_1 n_2 \cdots n_k$.

Problem 4. Find all solutions (when they exist) of the following linear congruences:

(i) $3x \equiv 1 \pmod{12}$;

(ii) $3x \equiv 1 \pmod{11}$;

(iii) $15x \equiv 5 \pmod{17}$;

(iv) $15x \equiv 5 \pmod{18}$;

(v) $15x \equiv 5 \pmod{100}$;

Problem 5. Solve the following systems of simultaneous linear congruences:

(i) $x \equiv 4 \pmod{24}$ and $x \equiv 7 \pmod{11}$;

(ii) $x \equiv 3 \pmod{5}$, $\quad x \equiv 1 \pmod{7}$, $\quad$ and $x \equiv 4 \pmod{8}$.